Experiment No-11

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# K-Means Clustering Algorithm

import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import make_blobs

# Step 1: Create a sample dataset
X, y = make_blobs(
    n_samples=300,
    centers=3,
    n_features=2,
    cluster_std=1.0,
    random_state=42
)

# Step 2: Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Step 3: Apply K-Means clustering
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

kmeans.fit(X_scaled)

# Step 4: Get cluster labels and centroids
labels = kmeans.labels_
centroids = kmeans.cluster_centers_
```

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# Step 5: Display cluster information
print("Cluster labels for the first 10 data points:")
print(labels[:10])

print("\nCluster centroids:")
print(centroids)

print("\nInertia:", kmeans.inertia_)
print("Silhouette score:",
    __import__('sklearn.metrics', fromlist=['silhouette_score'])
    .silhouette_score(X_scaled, labels))

# Step 6: Visualize the clusters
plt.figure(figsize=(8, 5))

plt.scatter(
   X_scaled[:, 0],
   X_scaled[:, 1],
   c=labels,
   cmap="viridis",
   s=40
)

plt.scatter(
   centroids[:, 0],
   centroids[:, 1],
   marker="X",
   s=200,
   color="red",
   label="Centroids"
)

plt.title("K-Means Clustering")
plt.xlabel("Feature 1 (scaled)")
plt.ylabel("Feature 2 (scaled)")
plt.legend()
plt.show()
```

Output of the code:-

```
Cluster labels for the first 10 data points:
[1 1 0 2 1 2 0 2 0 0]

Cluster centroids:
[[-0.21277029  1.15926337]
 [-1.07970061 -1.25787405]
 [ 1.2924709   0.09861068]]

Inertia: 18.26315933089595
Silhouette score: 0.8482330810203894
```

